

Shuai Yuan

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Education

- Master in Computer Science, National Tsing Hua University, Taiwan, China, 2014 (GPA: 4.23/4.3).
- Bachelor in Computer Science and Technology, Zhejiang University, China, 2012 (GPA: 3.84/4.0).

Research Experiences

- Lab member of LSA (Large-scale System Architecture) Lab, National Tsing Hua University, 2012–2014: work in the field of Cloud Storage System, Erasure Codes and GPGPU under the supervision of Prof. Jerry Chou. I was the TA of the CUDA lab class in my supervisor’s “Parallel Programming” course.
- Research intern in LRI (Laboratoire de Recherche en Informatique) of the University of Paris XI, France, 2011,10–2012,4: work on “automated constraint verification for databases” under the guidance of Prof.Véronique Benzaken and Prof. Évelyne Contejean.

Career Experiences

- Staff Engineer, NPU Department, ARM China Company, 2020/8–current.
 - Implement operators like Pooling, Matmul, RMSNorm, Reduce, RNN, etc.
 - Design and implement lowering passes of NN (Neural Network) compiler.
 - Performance optimization on multiple neural networks, including but not limited to:
 - * Optimize Reduce L2 in CodeFormer and achieve 4x speedup for all the Reduce L2 layer, cutting the overall network time by 9%.
 - * Optimize dynamic shape RMSNorm in Qwen and boost the performance by 68% for all the layers of the same kind in this network.
 - * Optimize RMSNorm in llama2 and achieve 16x speedup for all the layers of the same kind.
 - * Optimize Reduce SUM/MEAN in BEVFormer and achieve 6x speedup for all the layers of the same kind.
- Senior Software Engineer, Dorabot Company, 2018/10–2020/8.
 - I’m mainly working on collision detection, robot motion control, and performance optimization related to robot motion planning. My contributions include, but are not limited to:
 - * Performance optimization on collision detection for arm motion planning. I have achieved 6x speedup compared to the original implementation.

- * Maintain our fork of flexible collision library FCL. Use **SSE** and **AVX** to accelerate BVH overlap checking. My patch can improve the performance of OBB overlap checking by 30-60% compared to the latest version of FCL.
 - * Implement collision detection feature for grasp planning to support checking whether the moving gripper is collided with the obstacles in the environment. Use **OpenMP** to accelerate point cloud and octree processing.
 - * Implement driver for industrial robots like yaskawa motoman, fanuc, elfin, etc..
 - * Develop logistics induction application using **ros2 (Robot Operating System)**.
 - * Refactor RGBD camera driver and implement driver for Kinect2.
 - * Help vision colleagues with CUDA related issues and implement some of their customized algorithm in CUDA to achieve better performance.
- Heterogeneous Parallel Computing Engineer, Unicorn Technology, 2017/7–2018/9.
 - My contributions in Glass Flaw Detecting Project:
 - * Parallelize previous phone-glass flaw detecting implementation from 10s+ per glass to 1.2s.
 - * Implement GPU version of car-glass flaw detecting. We have achieved 5-6s for detecting flaws for each car glass.
 - * Accelerating Connected Component Labeling using **CUDA**.
 - * Optimize Morphology Erode/Dilate. In our cases, my implementation have gained 3-8 speedup compared to Nvidia NPP library.
 - My contributions in FPGA Smart Camera Project:
 - * Implement fixed-point quantization of Convolutional Neural Networks in C/C++ for verification.
 - My contributions in OCR (Optical Character Recognition) Project:
 - * Porting the OCR application to Nvidia Jetson **TK1** and **TX1**.
 - * Optimize mini-caffe deep learning framework in **TK1** and **TX1**.
 - My contributions in Mobile Face Detecting Project:
 - * Optimize YUV/RGB conversion, rotation etc. in **RenderScript**.
 - Mobile game developer in Xinyoudi Studio, Leqee Company, 2014/7–2017/5.
 - Join a startup team and develop a 2D mobile game (now available on **Google Play** and **App Store**).
 - As a leading client developer, I get my hands dirty on development using **cocos2d-x** and **quick-cocos2d-x**. Besides, I also take part in server development using **OpenResty**.
 - My main contributions include, but are not limited to:
 - * Trace errors and crash, work on fixing or workarounding upstream bug, some of the patches are accepted by quick-cocos2d-x (**PR#407**, **PR#438**, **PR#440**).
 - * Customize UI widgets.
 - * Write **Bash** or **Python** scripts for automate work flow, sorting and distribution of errors and crashes, server API press test, etc.
 - * Develop multiple modules in our game.
 - * Use **Docker** for quick deployment and scaling up.

Project Experiences

- GPU-RSCode: a GPGPU approach to accelerating Reed-Solomon codes for fault-tolerance in RAID-like system, 2012/12–2014/5.
 - Written in **CUDA C/C++**. Source code and documents are available under GPLv3: <https://github.com/yszbeda/GPU-RSCode>
 - We present an optimized GPU implementation of Reed-Solomon Codes, which can achieve a speedup of 14.71 over the current best CPU implementation Jerasure.

I have made some contributions to the following open-source projects:

- Embedded DDS network client **Micro XRCE-DDS Client**: fix udp multicast service discovery (PR #119, accepted).
- ROS-Industrial motoman packages: fix socket thread safety bug PR#319
- Linear algebra and geometry manipulation library MathGeoLib: fix SSE bug (PR#50, accepted).
- Multi-platform game framework: **cocos2d-x** and **quick-cocos2d-x**.
- Octopress plugin: octopress-syncPost (PR#8 & PR#9, one accepted).
- Utilities for nginx module development **openresty-devel-utils** (a simple patch PR#9 for lua-releng, accepted).
- Python wrapper for extended file system attributes: **xattr** (a simple patch PR#8, accepted).

Skills

- Programming Language: **Lua, C, C++, Bash**, etc.
- Framework/API: **CUDA, MPI, ros2, cocos2d-x, quick-cocos2d-x, OpenResty**, etc.
- IDE/Programming Tools: **XCode, Eclipse**, etc. (I use **Vim** as an editor, use **automake** to build my own project, have experience with **gdb, valgrind, gprof, cuda-gdb, cuda-memcheck, nvprof, adb, ndk-stack**, etc.)
- Operating System: **GNU/Linux** (Ubuntu, Debian, now I'm using ArchLinux).
- Version Control Tools: **git, svn** (As a git fan, now I prefer **git-svn** instead XD).
- Documentation: **L^AT_EX**

Honor and Awards

- Honor:
 - Navigator winner in Arm China Company, 2022 Q1.
 - Bravo Award in Tunicorn Technology Company, 2017–2018.
 - outstanding employee in Leqee Company, 2014–2015.
 - outstanding student in Zhejiang University, 2008–2009, 2009–2010, 2010–2011.
- Scholarship:

- Hong Hai/Foxconn scholarship, 2012–2013.
- Second prize of outstanding student scholarship and scholarship for academic, 2010–2011.
- Third prize of outstanding student scholarship and scholarship for academic, 2009–2010.
- Second prize of outstanding student scholarship and scholarship for academic, 2008–2009.
- Awards:
 - Outstanding thesis paper among undergraduate students, 2012.
 - Second prize of Zhejiang University-Intel Embedded Online Contest, 2010.
 - Third prize of Zhejiang University ACM Programming Contest, 2009.
 - Second prize of Zhejiang Province Calculus Contest, 2009.

Interests

- violin, Classical music, badminton, hiking, table tennis